האוניברסיטה הפתוחה
המחלקה למתמטיקה ומדעי המחשב

הצעה לתזה

חיזוי זמן הגעה משוער המבוסס על מודל רשת נוירונים גרפי דינמי מודע-מסלול

שם המנחה : פרופסור אהוד גודס
שם הסטודנט : גיא תורג'מן ת.ז. : 036238772

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תקציר

חיזוי זמן הגעה משוער (ETA) הוא מרכיב חיוני במערכות ניווט ותחבורה חכמה, אך תחזיות רבות אינן יציבות משום שהן נשענות על מצב התנועה הנוכחי בלבד. בפועל, עומסי תנועה מתפתחים לאורך זמן, וומודלים החסרים מנגנון לחיזוי השינויים בזרימת התנועה נוטים לספק תחזיות לא יציבות, המאופיינות בתדירות עדכון גבוהה ובסטיות ניכרות במהלך הנסיעה.

מרבית הגישות הקיימות מייצגות את התנועה ברמת הכביש או הצומת, מתעלמות מהאינטראקציות בין כלי רכב ואינן משלבות מידע מפורש על מסלול הנסיעה. עקב כך, הן מתקשות לאתר עומסים עתידיים הנוצרים מהצטברות רכבים לאורך מסלולים משותפים ומתכנסות לאזורי צוואר-בקבוק בתשתית.

מחקר זה מציע גישה חדשה לניבוי ETA המבוססת על שילוב ייחודי של מודל מודע-מסלול (Route-Aware) עם גרף תנועה דינמי ברמת הרכב. בייצוג זה, רשת הכבישים מורחבת כך שגם כלי הרכב עצמם מהווים צמתים בגרף, והקשרים ביניהם משתנים בזמן אמת בהתאם למיקומם, למהירותם, ולמסלולים שאליהם הם מתקדמים. שילוב זה מאפשר למודל לזהות תהליכי היווצרות עומס בטרם מתרחשים בפועל, באמצעות ניתוח דפוסי תנועה והצטברות כלי רכב במקטעי מסלול עתידיים משותפים.

הארכיטקטורה המוצעת (DSTRA-GNN) עושה שימוש משולב במידע מסלולי מפורש ובגרף תנועה דינמי ברמת הרכב, על מנת לתאר באופן מדויק את הדינמיות המרחבית-זמנית של זרימת התנועה. תוצאות ניסוי ראשוניות מבוססות סימולציה מצביעות על הפחתה משמעותית בשגיאת ה-ETA, עם MAE של 46.2 שניות — המהווה שיפור של כ-82% ביחס לקו בסיס ממוצע. שלבי המחקר הבאים יתמקדו בהרחבת המודל ליישום על נתוני אמת, בניתוח תרומתם העצמאית של רכיבי המערכת (Vehicle-Level Graph ו-Route-Aware), ובהערכת ביצועיו מול מודלי ETA מובילים משלוש פרדיגמות מרכזיות: Trajectory-based, Graph-based ו-Route-aware.

למחקר זה תרומה תאורטית ומעשית משמעותית. הוא מציג מסגרת אחודה להערכת מודלי ETA, מנתח באופן שיטתי את יעילותם של מודלים גרפיים דינמיים מודעי-מסלול על נתוני אמת, ומספק תובנות יישומיות הרלוונטיות למערכות ניווט, שירותי תחבורה, לוגיסטיקה וניהול ציים.

1 Work Objective

This thesis aims to improve ETA prediction accuracy by developing a dynamic Graph Neural Network (GNN) architecture that models traffic at the vehicle level with explicit route information. Unlike existing approaches using static sensor networks or implicit route inference, this work proposes a hybrid graph structure combining static infrastructure with dynamic vehicle nodes and time-varying edges. The primary research objective is to investigate whether explicit route information significantly enhances ETA prediction accuracy, comparing the proposed approach against baseline models including trajectory-based (DeepTTE [1], STAD [2]), static graph (DCRNN [3], ST-GCN [4]), and route-aware (DuETA [5]) methods.

2 Background with Reference to Sources

2.1 The ETA Prediction Challenge

Estimated Time of Arrival (ETA) prediction is a fundamental problem in intelligent transportation systems, requiring accurate prediction of remaining travel time given a vehicle's current position, planned route, and dynamic traffic network state. Unlike static shortest-path algorithms assuming constant travel times, ETA prediction must account for evolving traffic conditions, congestion buildup, and vehicle interactions.

Figure 1 illustrates a key challenge: traffic conditions change continuously as vehicles move through the network. The figure shows two snapshots (07:30 AM and 07:40 AM) tracking two vehicles traveling from a city to their homes. At 07:30 AM, Car 1's ETA is predicted as 07:55 AM. However, multiple vehicles at point A heading toward point B indicate potential future congestion. Ten minutes later at 07:40 AM, these vehicles reach point B, creating congestion that increases travel time from City to B from 15 to 35 minutes, updating Car 1's ETA to 08:15 AM—a 20-minute increase.

This example demonstrates the core research challenge: **how to accurately predict ETAs by anticipating future traffic conditions and congestion buildup, rather than relying solely on current traffic state**. Traditional navigation systems using real-time traffic data fail to account for traffic evolution during the journey, leading to inaccurate predictions requiring continuous updates. The research objective is to develop models that predict future congestion patterns and incorporate this information into ETA calculations, enabling more accurate and stable predictions.

2.2 Previous Work in Route-Aware Traffic Prediction

This research builds upon previous work in traffic prediction and route-aware modeling. Voloch and Voloch-Bloch [7] demonstrated real-time future traffic estimations for navigation route optimization, establishing the foundation for dynamic traffic prediction. Voloch et al. [6]

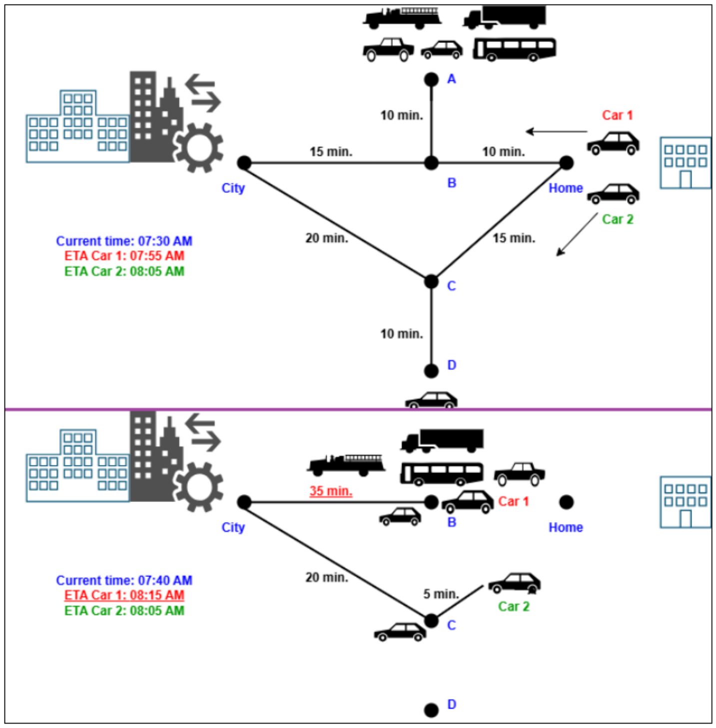

Figure 1: Illustration of the ETA prediction challenge: traffic conditions evolve dynamically, causing ETA predictions to change as congestion builds up. Figure adapted from [6].

introduced smart simulative route predictions that anticipate future congestion through heuristic simulation, improving ETA accuracy. These works laid the methodological groundwork for graph-based analysis of traffic dynamics and route-aware modeling.

2.3 State-of-the-Art ETA Prediction Methods

Machine learning models have been applied to ETA prediction with varying approaches. Tree-based methods such as XGBoost [8] leverage handcrafted features and perform well on aggregated trip records, but cannot capture fine-grained spatio-temporal interactions. Trajectory-based neural models address this gap: DeepTTE [1] learns ETA from raw GPS traces, while STAD [2] adjusts routing-engine outputs with spatio-temporal corrections. Production systems such as STANN [9] evaluate on large ride-hailing ETA datasets.

Graph-based spatio-temporal learning has advanced traffic prediction by representing road networks as graphs. DCRNN [3] combined diffusion graph convolutions with recurrent units, becoming a common backbone for traffic speed/flow prediction on sensor benchmarks (METR-LA [10], PEMS-BAY [11]). ST-GCN [4] introduced spatio-temporal graph convolutional networks, while Graph WaveNet [12] advanced deep spatial-temporal graph modeling. Most recently, DuETA [5] introduced duration-aware ETA modeling at Baidu Maps, categorizing trips into short (0–3 km), medium (3–10 km), and long (>10 km) segments, achieving MAEs of 27s, 46s, and 98s respectively. However, DuETA focuses on aggregated road segments rather than individual vehicle trajectories with explicit route intent.

2.4 Limitations of Existing Approaches

Existing benchmarks (NYC [13], Porto [14], Chengdu/DiDi [15], Geolife [16]) lack explicit representation of pre-planned routes, requiring models to infer paths between origin and destination, introducing ambiguity. Most ETA prediction methods either use static sensor networks (DCRNN [3], ST-GCN [4]) that aggregate traffic at fixed locations, or rely on trajectory-based learning (DeepTTE [1], STAD [2]) that infers routes implicitly.

Existing datasets present significant constraints: public datasets (NYC Taxi [13], Porto Taxi [14]) lack explicit route information and graph structures, while proprietary datasets (e.g., Baidu Maps used by DuETA [5]) are not publicly accessible. The absence of a unified dataset creates a fundamental challenge: trajectory-based models (DeepTTE [1], STAD [2]) require GPS trajectory sequences, static graph models (DCRNN [3], ST-GCN [4]) require aggregated sensor network data, and route-aware models (DuETA [5]) require route segment information. No existing public dataset provides the combination of explicit routes, dynamic graph structure, and vehicle-level data required for evaluating route-aware dynamic graph models, necessitating enhancement of real-world datasets with missing structural features while preserving ground truth travel times.

These limitations highlight the need for approaches that: (1) model traffic at the vehicle level rather than aggregated sensor networks, (2) explicitly encode predefined routes rather than inferring them implicitly, (3) combine dynamic graph structures with temporal learning to capture evolving traffic conditions, and (4) enable evaluation on real-world data by enhancing existing datasets with necessary structural features while maintaining statistical validity and ground truth.

3 Work Description

3.1 What Has Already Been Done

The DSTRA-GNN architecture has been implemented and evaluated on a four-week simulated SUMO dataset, achieving a mean absolute error (MAE) of 46.2 seconds, representing an 82.3% improvement over the average baseline (260.6 seconds MAE). These initial results provide evidence for the effectiveness of the route-aware dynamic graph neural network approach for ETA prediction.

The current implementation includes: (1) complete DSTRA-GNN architecture with dynamic graph representation, route encoding, and temporal learning mechanisms, (2) evaluation framework tested on simulated SUMO traffic data, (3) initial comparison showing significant improvement over average baseline models, and (4) four-week simulated dataset with dynamic graph snapshots at 30-second intervals.

While these results demonstrate the approach's potential, they are based on simulated data. The next phase will focus on validating the model's effectiveness on real-world data and establishing fair comparison with state-of-the-art baseline methods.

3.2 Planned Research Work

The planned research work is organized into three main phases, building upon the foundation established in the initial implementation:

3.2.1 Phase 1: Real Data Enhancement and Ground Truth Establishment

To validate the DSTRA-GNN model on real-world data, existing benchmark datasets (NYC Taxi, Porto Taxi, or similar) will be enhanced with additional features required by the DSTRA-GNN architecture. These datasets typically provide trip-level records with origin, destination, timestamps, and travel times, but lack explicit route information, dynamic graph structures, and vehicle-level interactions that DSTRA-GNN requires.

The enhancement process will add missing features in a statistically valid manner using controlled randomness and normal distributions that preserve the statistical properties of the original real data. This approach is critical because: (1) it maintains ground truth travel times from real-world observations, ensuring model evaluation reflects actual traffic conditions, (2) it preserves the statistical distribution and temporal patterns of the original dataset, (3) it enables fair comparison with baseline models using original dataset features while DSTRA-GNN benefits from enhanced features, and (4) it provides a common evaluation framework where all models are tested on the same underlying real-world traffic patterns, with ground truth established from actual observed travel times.

This approach preserves the real-world trajectories and travel-time ground truth, while enriching the dataset with additional contextual features generated from empirically calibrated statistical distributions to ensure completeness and consistency. The enhanced dataset will undergo comprehensive validation and statistical analysis to ensure that the original data properties are preserved, including distributional characteristics, temporal patterns, and ground truth travel times, before being approved for use in model evaluation.

3.2.2 Phase 2: Architecture Enhancements

Building upon the initial DSTRA-GNN implementation, this phase will explore architectural improvements to further enhance ETA prediction accuracy. Enhancements will focus on refining the model's ability to capture spatio-temporal traffic dynamics and route-aware patterns, determined through systematic analysis of the model's performance on the enhanced real-world dataset.

3.2.3 Phase 3: Comprehensive Baseline Comparison and Validation

Once the enhanced real-world dataset is available, comprehensive evaluation will be conducted comparing DSTRA-GNN against state-of-the-art baseline methods. This phase will: (1) implement and evaluate baseline models (DeepTTE, STAD, DCRNN, ST-GCN, DuETA) on the enhanced dataset, (2) establish performance benchmarks using standard metrics (MAE, RMSE, MAPE), (3) compare DSTRA-GNN performance against all baseline methods on the same real-world enhanced dataset, (4) conduct ablation studies to quantify the contribution of different architectural components, and (5) analyze performance across different trip characteristics and traffic conditions. This comprehensive evaluation will demonstrate whether the improvements observed on simulated data translate to real-world scenarios, providing stronger evidence for the effectiveness of the route-aware dynamic graph neural network approach.

3.3 Expected Deliverables

The work will produce: (1) an enhanced DSTRA-GNN architecture validated on real-world data, (2) a real-world dataset enhancement tool that adds necessary features while preserving statistical validity, (3) comprehensive experimental results comparing DSTRA-GNN against state-of-the-art baselines on real-world data, (4) detailed ablation studies quantifying architectural contributions, and (5) analysis demonstrating the effectiveness of route-aware dynamic graph modeling for ETA prediction in real-world scenarios.

4 Work Importance

This thesis addresses fundamental limitations in current ETA prediction research by developing a comprehensive evaluation framework and validating route-aware dynamic graph neural networks on real-world data.

Research Community Impact: Current ETA prediction research suffers from fragmentation, where different model architectures are evaluated on incompatible datasets, making fair comparison impossible. This thesis addresses this gap by developing a real-world dataset enhancement approach that enables rigorous evaluation of route-aware dynamic graph neural networks on actual traffic conditions while preserving ground truth travel times, providing the research community with a validated approach for evaluating advanced ETA prediction models.

Theoretical and Methodological Contribution: Initial results from the DSTRA-GNN model (detailed in Appendix A) on simulated data demonstrate that explicit route information combined with vehicle-level dynamic graph modeling significantly improves ETA prediction accuracy (46.2 s MAE, 82.3% improvement over average baseline). This thesis will provide the first comprehensive validation of route-aware dynamic graph neural networks on enhanced real-world datasets, establishing whether theoretical advantages translate to practical improvements. The real-world dataset enhancement approach enables evaluation of advanced models on actual traffic conditions while preserving ground truth travel times, establishing a new paradigm combining real data realism with structural richness required by modern architectures.

Practical Value: Accurate ETA prediction directly impacts navigation systems, ride-sharing platforms, fleet management, and logistics operations. By validating route-aware dynamic graph models on real-world data, this research provides evidence-based guidance for deploying advanced ETA prediction systems in production environments, with potential benefits for millions of daily users of transportation services.

5 Methods for Work Execution

The work will involve literature review, software development (implementing baseline models and extending DSTRA-GNN), experimental design and execution, data analysis, and documentation.

Data Collection and Preprocessing: Existing benchmark datasets (NYC Taxi, Porto Taxi, or similar) will be enhanced with missing features (explicit routes, dynamic graph structures, vehicle-level interactions) using statistically valid methods that preserve the original data properties. Preprocessing includes chronological partitioning, feature normalization, temporal window construction ($H = 30$ snapshots), route sequence encoding, and target normalization.

Analysis Techniques: Performance evaluation using standard regression metrics (MAE, RMSE, MAPE) computed per-vehicle and aggregated across trip categories. Ablation studies will compare route-aware vs. route-agnostic variants, temporal vs. non-temporal variants, and dynamic vs. static graph structures. Statistical analysis includes confidence intervals, significance testing, and error analysis across scenarios.

Validation Methods: Experiments will follow rigorous protocols including chronological data partitioning, hyperparameter tuning with early stopping, multiple runs with random seeds (42, 43, 44) for reproducibility, and model selection based on lowest validation MAE.

Ground Truth and Evaluation: Ground truth will be the actual travel times from the original real-world dataset. The enhanced dataset maintains these real-world travel times while adding necessary structural features, enabling fair comparison where all models are evaluated against the same ground truth.

Baseline Comparison: Comprehensive comparison on the enhanced dataset, ensuring all models (DSTRA-GNN, DeepTTE, STAD, DCRNN, ST-GCN, DuETA) are evaluated on identical traffic patterns. All code and configurations will be documented for reproducibility.

6 Work Execution Timeline

The work will be executed over ten months, organized into three main research phases.

Month 1: Phase 1 - Real Data Enhancement Tool Development - Development of statistical methods to enhance existing benchmark datasets (NYC Taxi, Porto Taxi) with missing features while preserving statistical validity and ground truth travel times.

Month 2: Phase 1 - Enhanced Dataset Generation - Application of enhancement methods to generate the complete enhanced real-world dataset with all required features.

Month 3: Phase 1 - Dataset Validation and Statistical Approval - Comprehensive validation of the enhanced dataset, statistical analysis to ensure preservation of original data properties, and approval of the dataset for use in model evaluation.

Month 4: Phase 1 Completion and Phase 2 Initiation - Finalization of validated enhanced dataset. Begin architecture enhancement analysis and design.

Month 5: Phase 2 - Architecture Enhancements - Analysis of DSTRA-GNN performance and implementation of enhancements focusing on spatio-temporal traffic dynamics and route-aware patterns.

Month 6: Phase 2 Completion and Phase 3 Initiation - Completion of architectural enhancements and initial evaluation. Begin baseline model implementation.

Month 7: Phase 3 - Baseline Model Implementation and Evaluation - Implementation and evaluation of baseline models (DeepTTE [1], STAD [2], DCRNN [3], ST-GCN [4], DuETA [5]) on the enhanced dataset using standard metrics (MAE, RMSE, MAPE).

Month 8: Phase 3 Completion and Comprehensive Evaluation Initiation - Completion of baseline evaluations. Begin comprehensive comparison and ablation study design.

Month 9: Comprehensive Evaluation and Analysis - Comprehensive evaluation comparing DSTRA-GNN against all baseline methods. Ablation studies, performance analysis across trip characteristics, and statistical synthesis.

Month 10: Thesis Writing and Finalization - Writing thesis chapters, preparation of figures and tables, final review, editing, and submission preparation.

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A Previous Work: Dynamic Route-Aware Graph Neural Networks for Accurate ETA Prediction

This appendix presents the paper ”Dynamic Route-Aware Graph Neural Networks for Accurate ETA Prediction” by Tordjman and Voloch, which serves as the foundation for this thesis research. The paper was submitted to IEEE Transactions on Intelligent Transportation Systems. The full paper is included below.

Dynamic Route-Aware Graph Neural Networks for Accurate ETA Prediction

Guy Tordjman
Department of Mathematics and Computer Science
The Open University
University Road 1, Ra'anana, Israel
Email: guy.tordjman@openu.ac.il

Nadav Voloch
Department of Computer Engineering
Ruppin Academic Center
Emek Hefer, Israel
Email: nadavv@ruppin.ac.il

***Abstract*—Accurate Estimated Time of Arrival (ETA) prediction is a central challenge in intelligent transportation systems, navigation services, and mobility applications. Existing approaches based on static routing algorithms or trajectory-driven machine learning improve over simple baselines but often fail to account for drivers' intended routes and the dynamic evolution of traffic interactions. These limitations restrict their ability to deliver reliable estimates under congested and heterogeneous traffic conditions.This work presents a dynamic spatio-temporal graph neural network that integrates vehicle-level dynamics, junction states, and explicit route intent within a unified Mixture-of-Experts framework. The transportation network is represented as a multi-relational graph, combining static road edges with dynamic vehicle interactions and planned route segments. Temporal dependencies are modeled through sequential graph snapshots, enabling the system to capture both persistent infrastructure characteristics and evolving traffic behaviors. Evaluation on a large-scale simulated dataset shows that the proposed model reduces Mean Absolute Error (MAE) from a simple average-time baseline of 260 seconds to 46 seconds, corresponding to an 82% improvement. In contrast, representative trajectory- and graph-based baselines typically achieve only moderate gains in the range of 40–50%. These results underscore the importance of explicit route representation and dynamic spatio-temporal modeling for accurate and robust ETA prediction.**

***Index Terms*—ETA prediction, spatio-temporal graph neural networks, traffic forecasting, mixture of experts, intelligent transportation systems.**

I. Introduction

Estimated Time of Arrival (ETA) prediction is a fundamental component of modern navigation systems and intelligent transportation applications. Accurate ETA enables commuters to make informed travel decisions, supports fleet management and logistics operations, and reduces congestion by distributing demand across the road network. With the ubiquity of GPS-enabled devices and connected vehicles, navigation platforms such as Google Maps and Waze [1]–[3] have transformed how travelers plan and adapt their journeys. Despite this progress, ETA estimation remains challenging due to the dynamic and stochastic nature of urban traffic (see Fig. 1).

Traditional approaches rely on shortest-path algorithms such as Dijkstra's algorithm [4], which efficiently compute routes under static conditions but fail to capture evolving congestion or vehicle interactions. As a result, travel times obtained from such methods can diverge substantially from reality when conditions change during a trip.

With the growth of urban mobility datasets, machine-learning models have been applied to ETA prediction. Tree-based methods such as XGBoost [5] leverage handcrafted features and perform well on aggregated trip records, including NYC Taxi and Porto Taxi [6], [7]. Deep learning approaches improve accuracy by modeling temporal patterns and spatio-temporal context along a route. For example, DeepTTE learns ETA from raw GPS traces [8], TADNM incorporates transportation-mode awareness [9], MetaTTE applies meta-learning for cross-city generalization [10], and STAD corrects routing-engine outputs using spatio-temporal adjustments [11].

In parallel, graph-based spatio-temporal learning has advanced traffic prediction by representing road networks as graphs and capturing dynamics through diffusion or convolution operators (e.g., DCRNN, ST-GCN, Graph WaveNet) [12]–[14]. However, existing benchmarks such as NYC, Porto, Chengdu/DiDi, and Geolife [6], [7], [15], [16] lack explicit representation of pre-planned routes. Models must therefore infer likely paths between origin and destination, introducing ambiguity and reducing accuracy.

To address this limitation, we represent traffic as a dynamic, multi-relational graph in which static junction-to-junction road edges are complemented by time-varying vehicle and interaction edges (see Fig. 2). This formulation builds on our earlier research in traffic simulation and route-aware modeling [17], [18], which laid the methodological groundwork for graph-based analysis of traffic dynamics. In the present work, we advance this foundation by explicitly encoding each vehicle's remaining planned route (route intent) and by learning temporal context from stable road edges across recent history, while leveraging the complete dynamic graph at the final snapshot to predict per-vehicle ETA.

Results. On a four-week urban simulation, a simple average-time baseline attains 260.6 s MAE; our route-aware non-temporal variant achieves 58.4 s MAE, and the temporal route-aware model reaches 46.2 s MAE. These gains highlight the value of making route intent explicit and of learning temporal signals from stable road infrastructure rather than noisy, transient vehicle interactions.

Paper outline. We introduce the dynamic graph representation

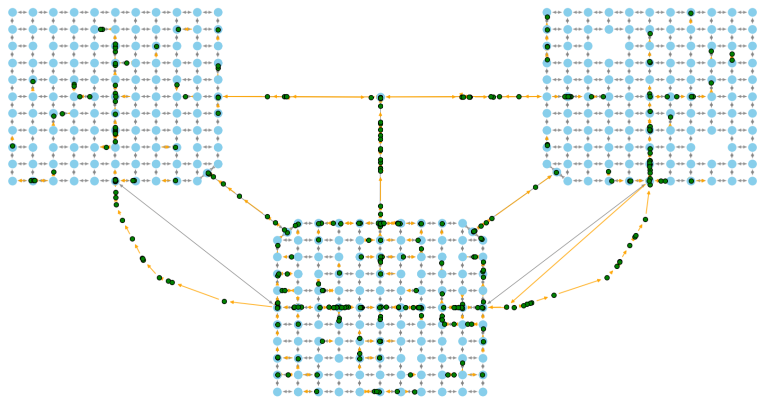


Fig. 1. Afternoon rush-hour snapshot spanning the network. Static junction nodes (light blue) and static road edges (gray) form the backbone; dynamic vehicle nodes (green) and traversal edges (orange arrows) reveal congestion corridors and high-flow funnels. Zones: A/B urban grids with dense signals and short links; C suburban ring with longer arterials and smoother flow; H connector hub that funnels inter-zone traffic and creates gateway bottlenecks.

and problem setup, summarize dataset characteristics and splits, detail the model architecture, and then present experiments and ablations followed by discussion and conclusions. To ground the setting early, Fig. 1 provides a system-level rush-hour snapshot.

Contributions. We introduce a dynamic, multi-relational traffic graph that unifies static junction nodes and road edges with dynamic vehicle nodes and interaction edges, and we provide explicit remaining-route annotations per vehicle to expose route intent. Building on this representation, we design a route-aware spatio-temporal GNN that learns temporal context from stable road edges over the first $H-1$ snapshots and fuses it with full-graph features at prediction time using a sparse Mixture-of-Experts head for specialization.

II. Related Work

A. Classical pathfinding and routing

On static graphs $G = (V, E)$ with non-negative edge costs, shortest paths are classically computed via Dijkstra's algorithm [4]. Alternatives such as Bellman–Ford [19] and bidirectional search [20] trade efficiency for generality. More advanced techniques such as A*/ALT landmarks [21], Contraction Hierarchies [22], and Customizable Route Planning [23] scale to web applications but rely on static edge-time models. As a result, they fail to capture evolving congestion or vehicle interactions. Our approach differs fundamentally by learning ETA directly from dynamic traffic graphs rather than static cost assumptions.

B. Learning-based ETA and traffic forecasting

The proliferation of urban mobility datasets motivated machine learning methods for ETA. Gradient boosting models such as XGBoost [5] leverage handcrafted features (trip distance, time-of-day) and perform well on aggregated taxi records (e.g., NYC, Porto [6], [7]), but they cannot capture fine-grained spatio-temporal interactions.

Trajectory- and trip-based neural models address this gap. DeepTTE [8] learns ETA from raw GPS traces and reports results on city-scale taxi/ride-hailing datasets (e.g., NYC TLC [6], Porto Taxi [7], DiDi 2016 [15]) with per-trip horizons typically within tens of minutes; MetaTTE [10] and related methods exploit cross-domain adaptation; STAD [11] adjusts routing outputs with spatio-temporal corrections. Production systems such as STANN [24] evaluate on large ride-hailing ETA datasets with short-to-medium trip durations (e.g., sub-30-minute trips). Complementary to these, the Smart Simulative Route framework [18] anticipates future congestion via heuristic simulation to improve ETA, but it remains simulation-driven and does not leverage graph-structured, data-driven temporal learning. Our work instead integrates explicit route intent and dynamic graph learning in a unified model.

C. Spatio-temporal graph neural networks

Graph neural networks advanced traffic forecasting by modeling sensor networks as spatio-temporal graphs. DCRNN [12] combined diffusion graph convolutions with recurrent units and became a common backbone for traffic speed/flow prediction on sensor benchmarks (METR-LA, PEMS-BAY; 5–60 min horizons in speed units). Notably, DCRNN-style encoders have also been adapted for ETA prediction in production settings, e.g., within DuETA [25]. In contrast to sensor-grid forecasting, we target per-vehicle ETA on dynamic multi-relational graphs with explicit route intent, while ConSTGAT [26] targets sensor-grid speed forecasting on METR-LA/PEMS-BAY at fixed horizons (not per-trip ETA).

Importantly, reported MAEs across these works are not directly comparable without dataset and time-horizon context (city, data source, and maximum trip duration). We therefore reference methods together with their evaluation scope rather than quoting unscoped headline numbers.

D. State-of-the-art production models

Most recently, DuETA [25] introduced duration-aware ETA modeling at Baidu Maps, categorizing trips into short (0–3 km), medium (3–10 km), and long (>10 km). DuETA achieved MAEs of 27s, 46s, and 98s respectively across these categories, significantly improving upon prior baselines. Unlike DuETA, which relies on trip segmentation, our method unifies all trip types under a single dynamic spatio-temporal framework. By incorporating route-left path features directly into the graph, our model can generalize across short and long trips without predefined bins. Moreover, we learn temporal context from stable road edges across history and fuse it at the prediction step, complementing DuETA's duration-aware design with explicit route intent and road-based temporal aggregation.

E. Summary

Prior work highlights the value of sequential modeling (DeepTTE), spatio-temporal attention (STANN, ConSTGAT), graph diffusion (DCRNN), and duration-aware embeddings (DuETA). However, none fully integrate explicit pre-planned routes with vehicle–junction dynamics within a unified dynamic graph. Our contribution is to combine explicit route intent, road-based temporal aggregation, and a sparse MoE head in a single spatio-temporal GNN for ETA.

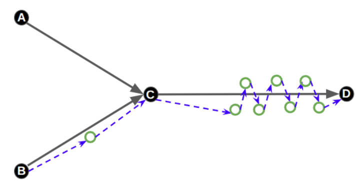
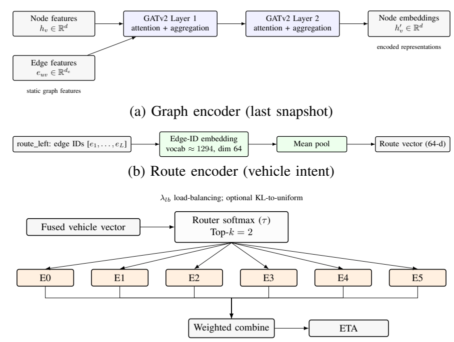


Fig. 2. Dynamic, multi-relational traffic graph: static junction nodes (labeled A–D) and road edges (solid gray) with dynamic vehicle nodes (hollow green circles) and interaction edges (blue dashed).



(a) Graph encoder (last snapshot)

(b) Route encoder (vehicle intent)

(c) Router and Mixture-of-Experts

Fig. 3. Key components: (a) two-layer GATv2 with edge features; (b) route intent via edge-ID embeddings and mean pooling; (c) sparse Top-$k = 2$ routing over six experts with load balancing.

III. Methodology

A. Dynamic Graph Representation

We represent the transportation system at snapshot time t as a directed multi-relational graph

$$G_t = \big(V_t, E_t, \{A_t^{(\rho)}, W_t^{(\rho)}\}_{\rho \in \mathcal{R}}\big), \tag{1}$$

where:

- $V_t = V^j \cup V_t^v$ is the set of nodes at time t, consisting of
  - V^j: static junction nodes, representing intersections in the road network,
  - V_t^v: dynamic vehicle nodes, representing vehicles present at time t.
- E_t is the set of directed edges at time t, partitioned by relation type ρ.
- $\mathcal{R}$ is the set of relation types (road, traversal, interaction).
- $A_t^{(\rho)} \in \{0,1\}^{|V_t| \times |V_t|}$ is the binary adjacency matrix for relation ρ at time t.
- $W_t^{(\rho)} \in \mathbb{R}_{\geq 0}^{|V_t| \times |V_t|}$ is the optional weighted adjacency for relation ρ (e.g., distance-based interaction weights).

We define three relation types:

1) **Road edges (ρ = road):** For adjacent junctions (u, v) with legal direction $u \to v$, we add $(u, v) \in E_t^{(\text{road})}$, encoding static road topology.
2) **Traversal edges (ρ = trav):** If vehicle v_i^t occupies segment (a, b), we add (a, v_i^t) and (v_i^t, b), linking the vehicle to its upstream and downstream junctions.
3) **Interaction edges (ρ = inter):** For vehicles v_i^t, v_j^t on the same segment with spacing $d_{ij}(t) \leq \varepsilon$ and aligned headings, we add (v_i^t, v_j^t) weighted by $\omega(d_{ij}(t)) = e^{-d_{ij}(t)/\lambda}$.

a) Dynamic edge construction.: In implementation, vehicles on each road segment are ordered by their normalized position along the edge. We then create a short chain per segment: a junction→first-vehicle edge (type 1), vehicle→vehicle links along the ordered list (type 3), and last-vehicle→junction (type 2). These dynamic edges sit on top of the static road graph (type 0).

Each node and edge carries feature vectors. Nodes use 28 features per snapshot (junction/vehicle type, kinematics, temporal encodings, zone one-hot, coordinates, current-edge attributes, demand/occupancy surrogates, and junction type). Edges use 7 features (average speed, lanes one-hot, length, demand, occupancy). Average speed, demand, and occupancy are updated per snapshot from data; vehicle features at indices for current-edge demand/occupancy are filled from the corresponding edge attributes.

b) Route intent.: In addition to graph-based relations, each vehicle node includes a feature called `vehicle_route_left`, which encodes the sequence of upcoming edges along its pre-computed source–destination path. Construction: for each trip we compute a static shortest path over E^{road} and record its ordered edge-ID list. At time t, we locate the vehicle's current edge within this list and take the suffix from the next edge to the destination. Indices are the canonical 0-based edge IDs shared across the dataset (vocabulary size 1,294). This sequence is summarized by a Route Encoder (edge-ID embedding + mean pooling) and used in fusion.

B. Temporal Windowing

ETA prediction requires reasoning over temporal dynamics. We construct a window of H consecutive snapshots:

$$\mathcal{G}_{t-H+1:t} = \{G_\tau\}_{\tau=t-H+1}^{t}, \tag{2}$$

where H is configurable (default H=30 in our experiments, with 30-second spacing). The prediction target is the ETA of vehicles present in the final snapshot G_t.

C. Problem Statement

Given a temporal window of dynamic graphs $\mathcal{G}_{t-H+1:t} = \{G_\tau\}_{\tau=t-H+1}^{t}$, where each snapshot G_τ contains static junction nodes V^j, dynamic vehicle nodes V_τ^v, static road edges, and time-varying interaction edges, the goal is to predict per-vehicle ETAs at the final snapshot $t^*=t$. Let $\mathcal{V}_t = V_t^v$ denote vehicles present at time t. We learn a function f_θ that maps

$$f_\theta\big(\mathcal{G}_{t-H+1:t},\ \texttt{route\_left}\big) \rightarrow \hat{\mathbf{y}}_t \in \mathbb{R}^{|\mathcal{V}_t|}, \tag{3}$$

where $\hat{\mathbf{y}}_t[i]$ is the ETA for vehicle $i \in \mathcal{V}_t$. Training minimizes mean absolute error (MAE) over vehicles present at t (additional terms are described in Section III-F):

$$\mathcal{L}_{\mathrm{MAE}} = \frac{1}{|\mathcal{V}_t|} \sum_{i \in \mathcal{V}_t} \big|\hat{y}_t[i] - y_t[i]\big|. \tag{4}$$

D. Model Architecture

Our DSTRA-GNN (Dynamic Spatio-Temporal Route-Aware GNN) integrates spatial, temporal, and route information. See the complete architecture in Fig. 4. The main components are:

- **Graph Encoder:** Each snapshot G_t is processed by a multi-layer GATv2-based encoder with residual connections and edge features, producing embeddings for junction and vehicle nodes (see the encoder panel in Fig. 3).
- **Temporal Aggregator:** For each pre-final snapshot $\tau \in \{t-H+1, \dots, t-1\}$ we take the static road-edge features (edge_type= 0) and build edge-wise sequences across time. A temporal module (Transformer with sinusoidal positional encodings; GRU optional) processes these sequences per edge; the temporally aggregated edge features are then projected and pooled over all static edges to a graph-level context vector (detail in Fig. 5). This context is broadcast to all vehicles at t^* for prediction. For *all* temporal variants, temporal context is computed from static road edges only for $t-H+1 \dots t-1$, while the final snapshot G_t is encoded with the appropriate spatial topology (static or full dynamic) per variant.
- **Vehicle Selection:** From the final snapshot G_t, we retain only vehicle node embeddings, as these are the prediction targets for ETA.
- **Route Encoder:** Each vehicle's remaining route (`vehicle_route_left`) is embedded via an edge-ID lookup (vocabulary $|E^{\mathrm{road}}|$=1,294, embedding dim 64) followed by mean pooling over the variable-length sequence (see the route panel in Fig. 3). The pooled route vector is concatenated with the vehicle embedding and temporal context when route awareness is enabled.
- **Fusion and MoE Head:** Vehicle embeddings, enriched with route and temporal context, are fused through a feed-forward network and routed to a sparse Top-k Mixture-of-Experts (MoE) head (router panel in Fig. 3), where specialized experts capture heterogeneous traffic regimes. The output is the ETA prediction for each vehicle.

E. Model Configuration

Unless stated otherwise, experiments use the following settings:

- **Node/edge features**: 28 node features; 7 edge features. Temporal window H=30 consecutive snapshots.
- **Graph encoder**: 2-layer GATv2 with edge features, hidden size 128, 4 heads per layer, GELU activations, LayerNorm residuals, dropout 0.1.
- **Temporal module**: static-road edge sequences over the first $H-1$ snapshots. Default is a Transformer Encoder with n_{head}=4, n_{layers}=2, GELU, FFN size $2d$, dropout 0.1, and sinusoidal positional encodings; an optional single-layer GRU (hidden 128) is also supported. Aggregated edge context is projected, mean-pooled over static edges to a graph vector, then broadcast to vehicles.
- **Route encoder**: edge-ID embedding of size 64 with mean pooling over `vehicle_route_left` (vocabulary size 1294).
- **Fusion**: MLP (256 hidden with GELU, LayerNorm, dropout 0.1) to a 192-dim fused representation.
- **MoE head**: Top-k=2 sparse routing over 6 experts (dropout 0.1). Predictions are made for vehicles in the last snapshot only.
- **Target normalization**: we train in the $y_{log,z}$ space:

$$y_{log} = \log\big(1 + y_{sec}\big) \tag{5}$$

$$y_{log,z} = \frac{y_{log} - \mu}{\sigma} \tag{6}$$

  where y_{sec} is the raw ETA in seconds, and (μ, σ) are computed on the training set; metrics are reported after inverting to seconds.
- **Training**: AdamW (lr 10^{-3}, weight decay 10^{-2}), linear warmup then cosine annealing ($T_{\max}$=200), batch size 2,

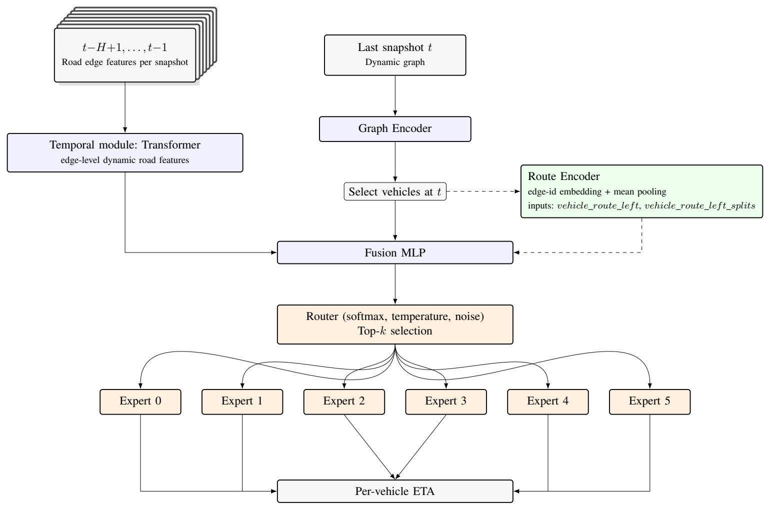


Fig. 4. DSTRA-GNN architecture. A window of T=30 dynamic graph snapshots (shown as an overlapped deck) is processed by a *shared* Graph Encoder; prediction is made at the last snapshot t^*. In the Full variant, a Route Encoder summarizes each vehicle's remaining path before Fusion and a Top-k MoE head produces per-vehicle ETA.

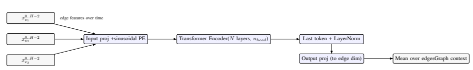


Fig. 5. Temporal Transformer detail: per-edge sequences across $H-1$ snapshots are projected, positionally encoded, encoded by a Transformer stack, reduced via last-token + norm, projected back to edge dimension, then pooled over edges.

200 epochs. Primary loss in target space with metrics reported after inversion to seconds.

- **Router regularization**: load-balancing weight λ_{lb}=0.05 plus optional KL-to-uniform router regularization with weight 0.002.

F. Loss Functions and Training Protocol

We train the model with supervised regression on ETA targets using mean absolute error (MAE) in seconds:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{N}\sum_{i=1}^{N}\left|\hat{y}_i - y_i\right|, \tag{7}$$

where $\hat{y}_i$ and y_i are the predicted and true ETAs. Additional evaluation metrics include RMSE and mean absolute percentage error (MAPE). The optimization objective includes MoE load balancing and optional router regularization: For completeness, we report the following definitions used in evaluation:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2} \tag{8}$$

$$\text{MAPE}(\%) = 100 \cdot \frac{1}{N}\sum_{i=1}^{N}\left|\frac{\hat{y}_i - y_i}{y_i}\right| \tag{9}$$

All metrics are computed in seconds (after inverting any target-space transforms) and averaged over vehicles in the test split.

$$\mathcal{L} = \mathcal{L}_{\text{reg}} + \lambda_{\text{lb}}\,\mathcal{L}_{\text{lb}} \;\; \text{(and optionally } \lambda_{\text{kl}}\,\text{KL}(p\,\|\,\text{Uniform})\text{)}, \tag{10}$$

where $\mathcal{L}_{\text{lb}}$ encourages uniform expert usage via importance/load terms from the router, and p are routing probabilities. Training occurs in a chosen target space (e.g., y_{log} or $y_{\text{log\_z}}$), while metrics are reported after consistent inversion to seconds.

a) Complexity and inference.: Per-snapshot spatial encoding is $\mathcal{O}(|E_t|)$; temporal aggregation is $\mathcal{O}(|E_{\text{road}}|\cdot(H-1))$ over static road edges only. Inference processes a window and emits per-vehicle ETAs at t^* with negligible overhead beyond a single forward pass of the encoder plus temporal aggregator.

The dataset spans four simulated weeks of traffic. We partition this chronologically into two weeks for training, one week for validation, and one week for testing. This split ensures that models are evaluated on non-overlapping time intervals that include different traffic patterns (rush hours, weekends, and long trips).

IV. Experimental Evaluation

A. Dataset

We evaluate on a simulation-derived dynamic traffic dataset built with the Eclipse SUMO traffic simulator [27], exported as time-ordered snapshots at 30-second intervals. Each snapshot $G_t = (V^j, V_t^v, E^{\text{road}}, E_t^{\text{dyn}})$ contains:

- **Static structure**: junction nodes V^j (types: traffic_light/priority) and directed road edges E^{road} (attributes: length, speed limit, num_lanes, zone).
- **Dynamic state**: vehicle nodes V_t^v with kinematics (speed, acceleration), positions, zone and current_edge; edge-time signals (avg_speed, vehicles_on_road_count, density).
- **Route-related features**: per-vehicle remaining route sequence (`vehicle_route_left`). For each trip, we compute the source–destination path using SUMO's Dijkstra implementation [4], [27] on the static road graph, with edge weights given by a travel-time cost derived from edge length and average speed. We then store the ordered list of edge IDs. At snapshot t, `vehicle_route_left` is the suffix of this list from the vehicle's current edge (exclusive) to destination; indices are 0-based edge IDs consistent with $|E^{\text{road}}|$=1,294. This sequence feeds the route encoder (edge-ID embedding + mean pooling) used by route-aware variants. We also include per-edge demand/occupancy proxies: `vehicles_on_road_count` and its log, and the number of routes that traverse each edge (`edge_route_count`, log).

Fig. 1 shows an afternoon rush-hour snapshot illustrating realistic congestion buildup across three zones: (i) **A/B urban grids** with dense signalized junctions and short links, driving stop-and-go dynamics and rich vehicle interactions; (ii) **C suburban ring** with longer arterials and fewer signals, yielding smoother flows; and (iii) **H hub/connector** that funnels inter-zone traffic, creating recurrent bottlenecks at gateways. The zone label is available at both node and edge level and conditions many dynamic features (e.g., average speed, occupancy), which the model can leverage.

Scope and scale:

- Duration and coverage: ∼4 weeks; snapshots every 30 s; total snapshots: 80,730.
- Topology: junctions: $|V^j|$=362; static road edges: $|E^{\text{road}}|$=1,294.
- Traffic: unique vehicles: 208,763; trips: 22,767,090; avg active vehicles per snapshot: 282.02.

Chronological splits are used throughout (no overlap): train, validation, and test. Unless otherwise stated, we use a temporal window of H=30 snapshots (15 minutes) and a stride of 10 between windows.

B. Dataset Statistics

Key feature statistics:

- **Vehicles**: mean speed 12.11 m/s (std 10.21), acceleration mean -0.15 m/s^2 (std 1.12); route length mean 13.07 km (median 13.98 km); remaining route (`route_length_left`) mean 7.60 km (median 7.90 km); vehicle types: passenger/truck/bus.
- **Junctions**: 362 nodes; types split between traffic_light and priority; zones A/B/C/H.
- **Edges**: speed limit median 13.89 m/s; length median 485.6 m; lanes in {1,2,3}; avg vehicles on road count is sparse (median 0), heavy-tailed; `edge_route_count` is highly skewed (max 785), use log scaling.
- **Labels/ETA**: ETA mean 504 s (std 416), spans 0–∼5000 s; total travel time mean 1011 s.

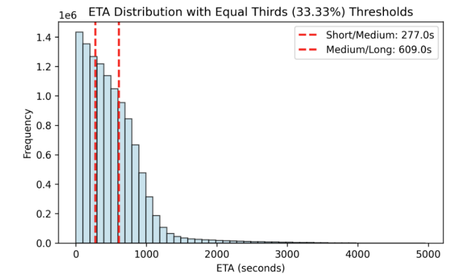


Fig. 6. ETA distribution (seconds) with equal-thirds thresholds. Vertical dashed lines indicate short/medium (≈277 s) and medium/long (≈609 s) boundaries.

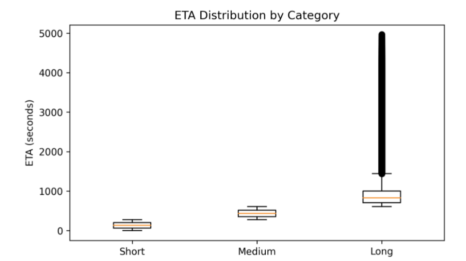


Fig. 7. ETA distribution by route-length category (short/medium/long). Boxes summarize medians and IQRs; whiskers capture spread and outliers.

ETA statistics and bin thresholds are shown in Figs. 6–8. We use equal-thirds thresholds (Fig. 6) at ≈277 s and ≈609 s to define short/medium/long duration bins; validation counts per bin are listed in Table III.

C. Experimental Setup

Models are trained using AdamW with learning rate 10^{-3} and weight decay 10^{-2}, batch size 2, for 200 epochs. The schedule uses a short linear warmup followed by cosine annealing ($T_{\max}$=200). Targets are learned in a normalized space (default $y_{\text{log\_z}}$), while metrics are computed after consistent inversion to seconds. The router uses load-balancing (weight 0.05) and KL-to-uniform regularization (weight 0.002). Unless stated otherwise, reported numbers are the mean over seeds 42, 43, and 44; model selection uses the lowest validation MAE per seed.

D. Baselines and Ablations

We compare six code-defined variants (shared encoder and training protocol). Table I summarizes the variants and which

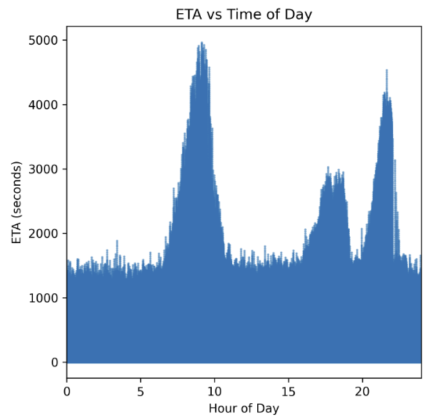


Fig. 8. ETA versus hour-of-day showing strong diurnal effects (morning and evening peaks), with occasional evening events/spikes.

TABLE I
ABLATION VARIANTS AND ENABLED COMPONENTS. TEMPORAL USES A TRANSFORMER WITH STATIC-ROAD CONTEXT FOR THE FIRST $H-1$ SNAPSHOTS.

Variant	Dyn. Edges	Route Feat.	Temporal
base_graph	—	—	—
dynamic_graph	✓	—	—
route_aware_graph	✓	✓	—
temporal_base	—	—	✓
temporal_dynamic	✓	—	✓
temporal_route_aware	✓	✓	✓

components are enabled:

- `base_graph`: static road graph only; no dynamic edges; no temporal; no route features.
- `dynamic_graph`: adds dynamic edges (junction→vehicle, vehicle→vehicle, vehicle→junction); no temporal; no route features.
- `route_aware_graph`: adds route features/encoder and dynamic edges; no temporal.
- `temporal_base`: temporal context from static-road edges only for the first $H-1$ snapshots; no route features; spatial encoder uses only static edges.
- `temporal_dynamic`: temporal context from the first $H-1$ snapshots is aggregated over static-road edges; spatial encoder uses dynamic edges; no route features.
- `temporal_route_aware`: full model with dynamic edges, route features/encoder, and temporal context aggregated from static-road edges for the first $H-1$ snapshots.

E. Evaluation Metrics

We report Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) in seconds, and Mean Absolute Percentage Error (MAPE). We also provide stratified MAE by route-length and duration bins.

TABLE II
OVERALL AND PER-BIN VALIDATION MAE (SECONDS) AVERAGED OVER SEEDS 42–44 (BEST EPOCH PER SEED). IMPROVEMENT IS RELATIVE TO THE AVG BASELINE.

Variant	MAE	Short	Medium	Long	Improve vs AVG
avg	260.63	282.20	83.56	430.67	0.0%
base_graph	86.46	39.20	81.79	140.27	66.8%
dynamic_graph	104.51	42.65	95.99	177.54	59.9%
route_aware_graph	58.42	24.19	54.84	97.61	77.6%
temporal_base	92.62	61.35	83.54	135.00	64.5%
temporal_dynamic	78.79	32.04	67.58	142.14	69.8%
temporal_route_aware	46.16	24.05	38.31	76.90	82.3%

V. RESULTS AND ANALYSIS

Validation results. Tables II and IV report the performance of all ablation variants. The naive average-time baseline (`avg`) yields 260.6 s MAE and 325.1 s RMSE, with a MAPE of 127.8%, confirming the task's difficulty.

The `base_graph` model, which uses only static road topology and basic vehicle kinematics, reduces MAE to 86.5 s (66.8% improvement). Short trips are predicted with 39.2 s MAE, while medium and long trips remain harder (81.8 s and 140.3 s).

Adding dynamic interaction edges in `dynamic_graph` achieves 104.5 s MAE (59.9%), weaker than the static-only model. Errors for medium/long trips (95.9 s and 177.5 s) indicate that dynamic edges without route features add noise rather than signal.

The `route_aware_graph` variant delivers strong gains: 58.4 s MAE overall, with 24.2 s (short), 54.8 s (medium), and 97.6 s (long). Explicitly encoding planned routes and edge demand/occupancy lets the model anticipate congestion along a vehicle's actual path.

Temporal variants further refine performance. `temporal_base` achieves 92.6 s MAE (64.5%), mainly improving medium trips but struggling on short trips (61.4 s). `temporal_dymanic` improves to 78.8 s MAE (69.8%), showing that temporal memory of interaction edges helps even without route features.

The best performance is `temporal_route_aware`, which combines temporal aggregation with explicit route features: 46.2 s MAE and 107.2 s RMSE (82.3% improvement), with balanced per-bin accuracy (24.1 s short, 38.3 s medium, 76.9 s long) and the lowest MAPE (17.9%).

Equally important is the underlying dataset representation. Unlike prior work that models only road segments or aggregated trajectories, our dataset is structured as a dynamic graph where both junctions and vehicles are nodes. Static road edges encode the persistent infrastructure, while dynamic edges evolve at every snapshot to reflect vehicle–junction and vehicle–vehicle relationships. This design allows the model to jointly learn from the stable backbone of the road network and the transient dynamics of traffic flow, enabling the strong gains observed in route-aware and temporal variants.

TABLE III
DURATION BINNING USED THROUGHOUT (VALIDATION SPLIT). THRESHOLDS FROM FIG. 6.

Bin	ETA range (s)	τ (s)	Count
Short	[0, 277]	30	33,227
Medium	(277, 609]	60	34,996
Long	> 609	120	32,228

TABLE IV
OVERALL VALIDATION MAE AND RMSE (SECONDS) AND MAPE (%). BEST EPOCH PER VARIANT ON VALIDATION; AVERAGED OVER SEEDS 42–44 WHERE AVAILABLE. MAPE VALUES ARE ESTIMATED WHERE MISSING IN LOGS.

Variant	MAE (s)	RMSE (s)	MAPE (%)
avg	260.63	325.13	127.75
base_graph	86.46	145.08	33.70
dynamic_graph	104.51	157.65	40.73
route_aware_graph	58.42	106.99	22.77
temporal_base	92.62	155.48	26.51
temporal_dynamic	78.79	127.88	30.70
temporal_route_aware	46.16	107.15	17.99

VI. DISCUSSION

The results highlight three clear trends. First, static road graphs already provide a strong inductive bias, with the `base_graph` achieving a two-thirds reduction in error relative to the average baseline. This demonstrates the value of graph-structured representations in capturing spatial constraints.

Second, route intent emerges as the most decisive feature. The `route_aware_graph` achieves 58.4 s MAE overall, more than 40% lower than either the `base_graph` or `dynamic_graph`. Notably, short trips drop from 39–43 s error to just 24 s, indicating that route features resolve much of the ambiguity about vehicle destination and path choice.

Third, temporal context provides an additional layer of refinement. The `temporal_route_aware` variant achieves 46.2 s MAE and 17.9% MAPE, corresponding to an 82.3% improvement over the average baseline and outperforming all other variants across every trip-length category. The temporal aggregation of $T-1$ snapshots enables the network to recognize evolving congestion patterns, improving robustness for medium and long trips where static and route-only models tend to accumulate error.

These gains are substantially higher than those reported by state-of-the-art methods on real-world datasets. For example, DuETA reports MAE reductions of 40–50% relative to average baselines across Beijing, Shanghai, and Tianjin [25], while earlier models such as DCRNN, DeepTTE, and ConSTGAT report similar ranges [8], [12], [26]. In contrast, our model delivers over 80% improvement, underscoring the benefit of combining explicit route intent with dynamic spatio-temporal graph modeling.

Together, these findings validate the dataset design, which provides per-vehicle route annotations, dynamic interaction edges, and temporally aligned graph snapshots. Each of these components is directly reflected in the ablation gains. At the same time, long trips remain the most challenging case (76.9 s MAE even in the best model), indicating that compounding local errors across extended horizons remains an open problem.

A further contribution of this work lies in the dataset formulation itself. By explicitly representing both junctions and vehicles as nodes, and layering dynamic edges on top of static road topology, the traffic network is captured in a richer and more flexible way than traditional road-segment–based graphs. This representation provides the inductive bias needed to integrate infrastructure constraints with real-time vehicle interactions, and the ablation results demonstrate that it is precisely this combination—static roads, dynamic edges, and route-aware features—that yields the largest performance gains.

The scale and temporal coverage of the dataset further reinforce these results. With over 80,000 graph snapshots, 200,000+ vehicles, and approximately 927,000 distinct routes across a continuous four-week horizon, the evaluation provides dense temporal coverage and rich route diversity. This extended duration ensures that models are exposed to recurring traffic cycles such as rush hours, weekday–weekend variations, and long-term congestion patterns, while the large number of routes captures heterogeneous travel intents and path choices. Compared to widely used benchmarks such as METR-LA (342 loop sensors, aggregated speeds over a few months) or trajectory datasets like Q-Traffic and DiDi, which provide trip-level GPS traces without explicit route intent or graph structure, our dataset offers far richer temporal, spatial, and behavioral signals. Importantly, this level of detail reflects information readily available in modern navigation systems through GPS traces, map matching, and planned-route logging, making the evaluation both realistic and robust while reducing the risk of overfitting to short-term artifacts.

VII. CONCLUSION

This work introduced a dynamic spatio-temporal graph neural network for ETA prediction that unifies static infrastructure, vehicle-level dynamics, and explicit route intent within a Mixture-of-Experts framework. A central innovation of our approach is the formulation of the traffic environment as a dynamic graph in which both junctions and vehicles are nodes, connected by persistent road edges and evolving interaction edges. This representation enables the model to leverage the stability of the road network while simultaneously adapting to transient traffic patterns.

By systematically evaluating ablation variants, we demonstrated that each component contributes to predictive accuracy: graph structure captures spatial constraints, route features provide the largest single boost, and temporal aggregation refines predictions by modeling evolving congestion. The resulting temporal route-aware model reduced validation MAE from 260.6 s in the average baseline to 46.2 s, an 82.3% improvement, with balanced accuracy across short, medium, and long trips. Compared with improvements typically reported in the range of 40–50% by state-of-the-art baselines, these results highlight the benefit of representing the traffic system as a dynamic vehicle–junction graph.

Future work will focus on scaling to real-world datasets, addressing error accumulation in long trips, and ensuring robustness under unexpected disruptions, thereby advancing the practical deployment of intent-aware dynamic graph models in intelligent transportation systems.

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